Credit Risk Portfolio Analytics — Solution Overview

Analyst workflow

Portfolio view tracks exposure, defaults and predicted risk across loan, region, employment and cross-dimensional cuts. Filters flow into all views. Account review shows score, segment and observed drivers. Model governance shows holdout performance, global importance and monitoring controls.

AI features & controls

Ask: submitting a supported question triggers a computed aggregation; the optional LLM only phrases that verified fact. FP: misunderstood filter/denominator; FN: valid but unsupported wording; analyst checks source and filters. **Anomaly:** ≥3 robust SD above loan/income peers triggers a reason-coded flag. FP: legitimate unusual customer; FN: deterioration within peer norms; analyst confirms/dismisses, with no automatic adverse action.

Model

Account-grain bureau, repayment, collections, customer and loan features; stratified 75/25 holdout; median/mode imputation fitted inside pipelines; logistic baseline and histogram gradient-boosting challenger. Selected: **logistic regression**. Holdout AUC **0.675**, KS **0.265**, Gini **0.350**. Monitor AUC/KS/Gini, calibration, PSI, missingness, fairness and review outcomes monthly.

Top portfolio insights

1) Mortgage has the highest default rate (24.7%); prioritize underwriting/collections review for that product. 2) Unemployed customers have the highest employment-segment default rate (35.0%); verify income stability and avoid blanket policy decisions. 3) South has the highest regional default rate (16.9%); investigate product/employment mix before changing limits.

Assumptions & production needs

Synthetic target and histories are treated as observation-safe; scores are decision support, not adverse-action automation. Production needs: governed source contracts and lineage, point-in-time feature store, identity/access controls, encrypted secrets, model registry/validation, audit logs and durable review database, consent/retention rules, bias and regulatory review, monitoring SLAs, incident rollback and bank-approved LLM/data-processing terms.